Xiangyu Ding

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Research interests

Integer partitions; q-series; Bailey chains and lattices; truncated theta series; Rogers–Ramanujan-type identities; representation theory of Lie algebras.

Education

Integrated graduate-to-PhD programme in Applied Mathematics, Nankai University | September 2021 – June 2027 (expected)

Doctoral stage since September 2023

Supervisor: Professor Lisa Hui Sun

Papers

[1] X. Ding and L. H. Sun, Truncated theta series from the Bailey lattice. *Advances in Applied Mathematics* 167 (2025), 102884. arXiv:2403.11608.

[2] X. Ding and L. H. Sun, A combinatorial proof for the positivity of the normalized Jacobi triple product tails. *Journal of Combinatorial Theory, Series A*, accepted. arXiv:2606.27507.

[3] X. Ding and L. H. Sun, On Merca's stronger conjecture for the truncated Jacobi triple product series. submitted. arXiv:2411.13818.

[4] I. D. Chen, X. Ding, Q. Guo and L. H. Sun, The extension of Ballantine and Merca's conjecture on 6-regular partitions. submitted.

[5] X. Ding and L. H. Sun, A bijective partition-theoretic interpretation for the Andrews-Merca truncated pentagonal number theorem. submitted.

[6] X. Ding and L. H. Sun, A symmetrized B_2 Bailey lemma and Rogers–Ramanujan-type identities. preprint.

Awards

Shiing-Shen Chern Academic Newcomer Scholarship, Nankai University, 2025

陈省身学术新人奖学金

Academic activities

Participant, 36th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC 2024), Ruhr-Universität Bochum, Germany, 22–26 July 2024.

Research methods

Bailey theory and finite transformations: Bailey chains and lattices, conjugate WP–Bailey pairs, and finite kernel invariance. Applications include Andrews–Gordon-type truncations and symmetrized B_2 Rogers–Ramanujan-type multisums.

Bijective methods: weight-preserving maps built from Durfee rectangles, staircase operations and Gaussian refactorization; sign-reversing involutions and generalized minimal-excludant classes. Frobenius-arm lifts provide injections proving bivariate coefficientwise nonnegativity for normalized Jacobi triple product tails.

Coefficient analysis: exact enumeration of restricted partitions and residue-class estimates, together with effective circle-method bounds for nonmodular infinite products. These methods yield explicit eventual-nonnegativity thresholds and inequalities for regular partitions.

Connections with representation theory: multisum expressions for selected principally specialized affine A_1 characters and extremal CMPP products, derived from the symmetrized B_2 framework.

References

Professor Lisa Hui Sun (PhD supervisor)
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